

Physiology: Organization of the Urinary System

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Session Objectives

At the conclusion of the session, the student should be able:

- 1. Describe the major functions of the kidneys.**
- 2. Define & describe the gross structures and their interrelationships: renal pelvis, renal pyramids, renal medulla (inner and outer zones), renal cortex, and papilla.**
- 3. Define & describe the components of the glomerular filtration barrier**
- 4. Define & describe the components of the nephron-collecting duct system and their interrelationships.**
- 5. Define & describe the juxtaglomerular apparatus and its role in homeostasis.**

You just started your rotation in Emergency Medicine when a 17-year-old is brought in to the ER by her parents. She is an avid soccer player and has no significant PMH. She was 167 cm tall, and her weight was 48 kg (BMI = 17 kg/m²). She is in acute distress and you find her curled up in a fetal position in her bed.

The patient was nauseous, and the PE revealed signs of dehydration as well as deep, bilateral flank pain. Diuresis was quantified as low (0.5L/day) and urinalysis showed microscopic hematuria (+++), proteinuria (+++), and glycosuria (+++) with normal glycemia.

After a conversation with her parents, you learn that she is extremely health-conscious and an avid social media user. They also tell you that she had gotten some herbal teas from a popular European spa and had just drunk some of it in the previous 24 hrs.

You wonder if there is a connection.

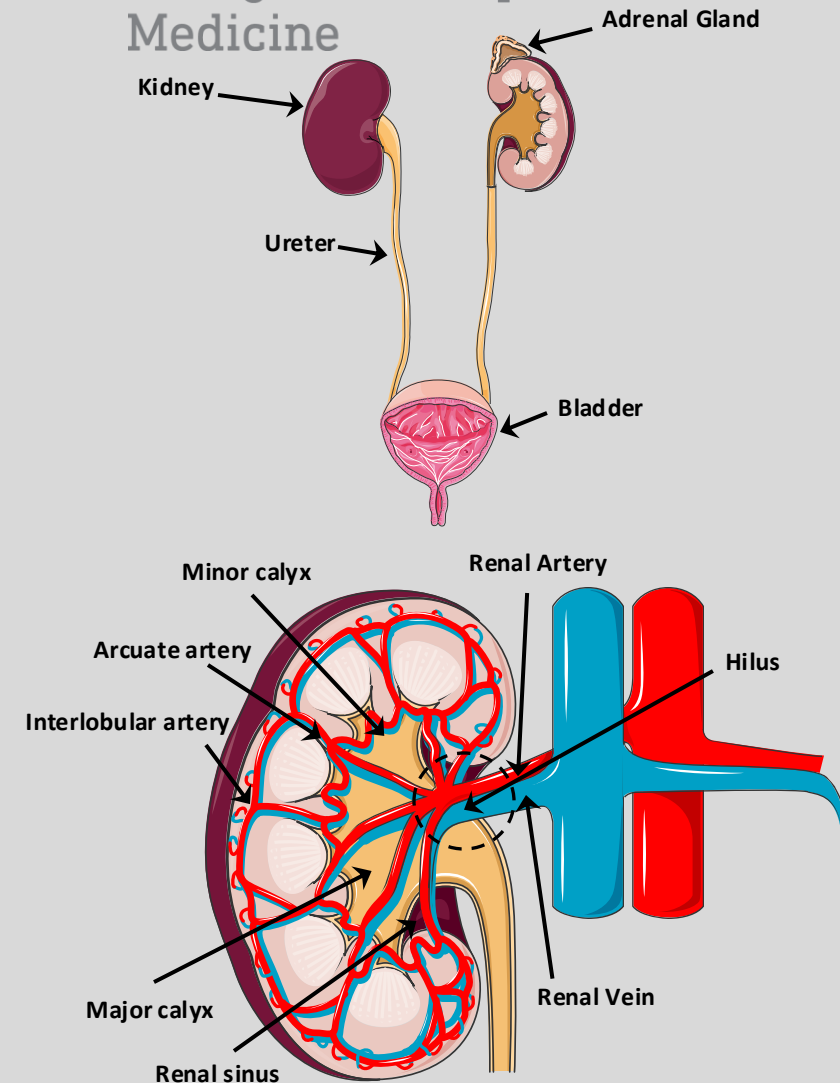


Functional Anatomy of the Kidney

- The kidneys serve three essential functions:
 - Filtration: Removes metabolic products & toxins from the blood & excretes them through the urine.
 - Regulation of fluid status, electrolyte balance & acid-base balance.
 - Production or activation of hormones involved in erythropoiesis, Ca^{++} metabolism, & regulation of blood pressure & flow.
- Paired, bean-shaped structures that lie behind the peritoneum on each side of the vertebral column (L12 to L3).
- Comprise less than 0.5% of the TBW
 - Males: 125-170 g each
 - Females: 115-155 g each.
- Hilus: Site of entry for the renal artery & nerves & exit for the renal vein.
- Renal artery: Enters the hilus and divides into anterior and posterior branches, which give rise to the arcuate and then interlobular arteries.
- Arcuate arteries:
 - Skirt the corticomedullary junction
 - Branch into ascending interlobular arteries → afferent arterioles.
- Renal sinus:
 - Includes the renal pelvis & the major and minor calyces (urine-filled spaces)
 - Surrounded by renal parenchyma except where it connects with the upper end of the ureter.

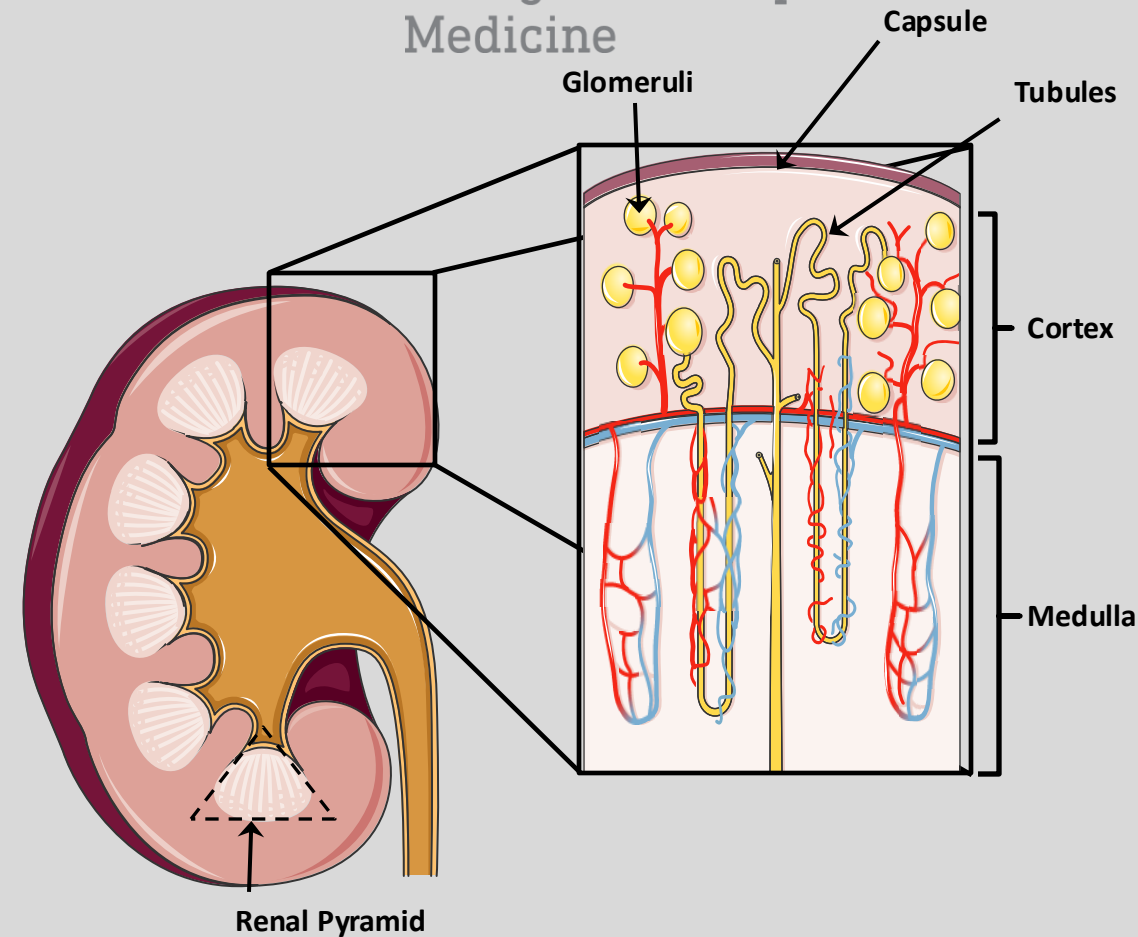
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Functional Anatomy of the Kidney

- A fibrous capsule covers each kidney.
- The renal capsule reflects into the sinus at the hilus
- Its inner layers line the sinus
- Outer layers anchor to the blood vessels and renal pelvis.
- Kidney cross-section reveals two visible layers:
 - **Cortex:**
 - Granular outer region due to the presence of glomeruli & highly convoluted tubules
 - **Medulla:**
 - A darker inner region.
 - Lacks glomeruli
 - Parallel arrangement of tubules and small blood vessels.
 - Subdivided into 8-18 conical renal pyramids
 - **Renal Pyramids :**
 - Base faces the cortical-medullary border
 - Tip terminates in the renal pelvis.



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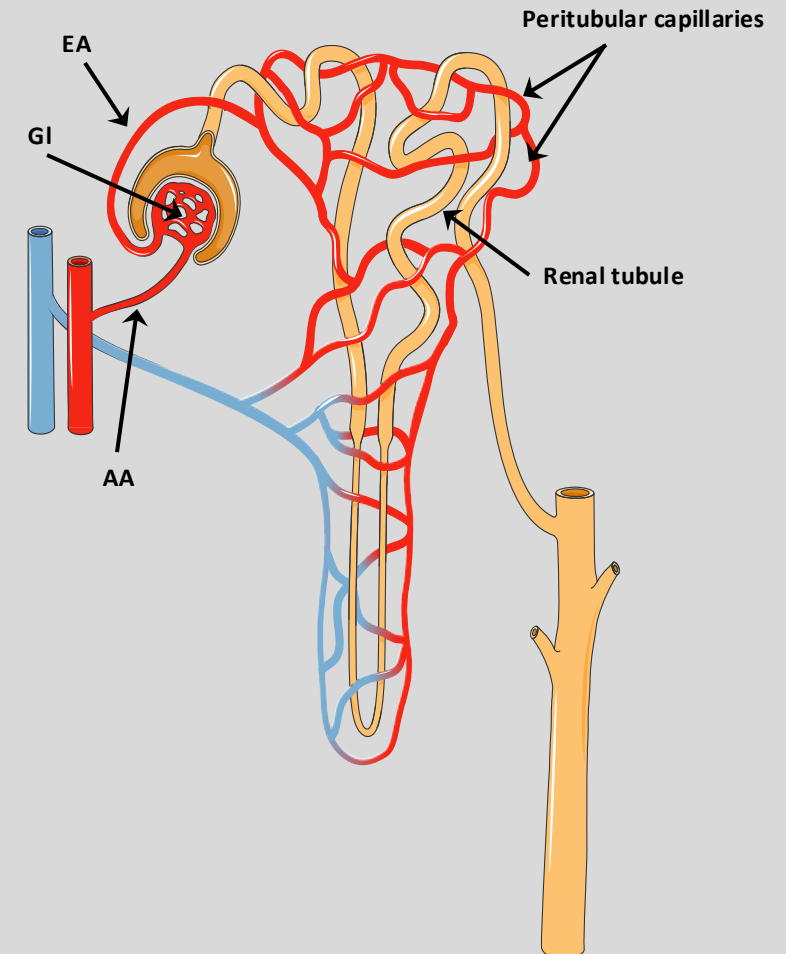
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Vasculature of the Kidney

- The kidneys have a very high blood flow and complex & unique vasculature
- They receive ~ 20% of the cardiac output.
- High blood flow provides the blood plasma necessary for forming ultrafiltrate
- Renal circulation:
 - Afferent arteriole (AA): High-resistance arteriole bringing blood into the glomeruli.
 - Glomerulus (GI): High-pressure capillary network for filtration
 - Efferent arteriole (EF): 2nd high-resistance arteriole
 - Peritubular capillaries: Low-pressure capillary network surrounding the renal tubules
- Juxtamedullary glomeruli:
 - Located at or near the medullary & cortical junction.
 - Efferent arterioles of these nephrons descend into the renal papillae to form hairpin-shaped vessels called the vasa recta
 - Vasa Recta: The capillary network for tubules in the medulla.
- ~90% of the blood entering the kidney perfuses superficial GL and cortex
- ~10% perfuses juxtamedullary GL and medulla.
- Lymph vessels:
 - Drain cortical ISF & contain renal hormones (e.g., erythropoietin)
 - Leave the kidney by following the arteries toward the hilus.
 - Absent from the renal medulla

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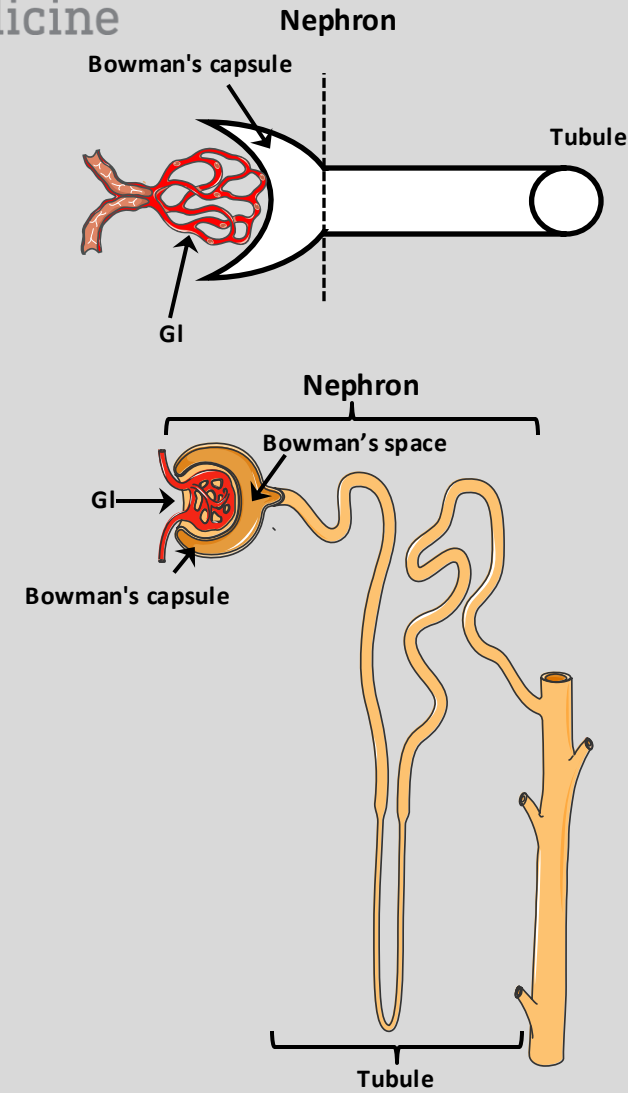


The Nephron

- The functional unit of the kidney is the nephron (800,000 to 1,200,000/kidney)
- Consists of a glomerulus and a tubule.
- Each nephron is an independent entity until the point at which its initial collecting tubule merges with another tubule.
- 2 populations of nephrons w/i the renal cortex:
 - Superficial nephrons: Have short loops extending to the boundary between outer and inner medulla.
 - Juxtamedullary nephrons: Play a special role in the production of concentrated urine with long loops that extend into medulla.
- Renal corpuscle:
 - The site of formation of the glomerular filtrate, comprises of the following:
 - Glomerulus: Cluster of capillaries from which the plasma filtrate originates.
 - Tubule: Epithelial structure consisting of subdivisions, designed to convert the filtrate into urine.
 - Bowman's capsule (glomerular capsule):
 - Epithelial structure surrounding the glomerulus
 - Contains Bowman's space
 - Contiguous with the lumen of the tubule.
 - Where the filtrate passes from the vascular into the tubule system.
- Bowman's space:
 - The space between parietal cells & glomerular capillaries

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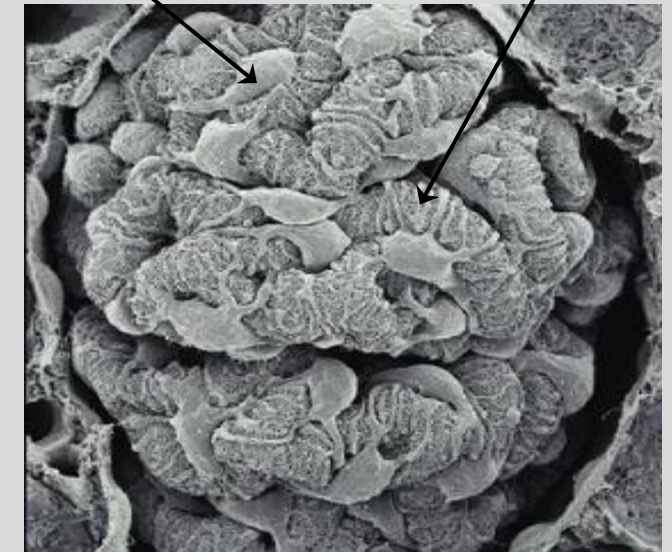
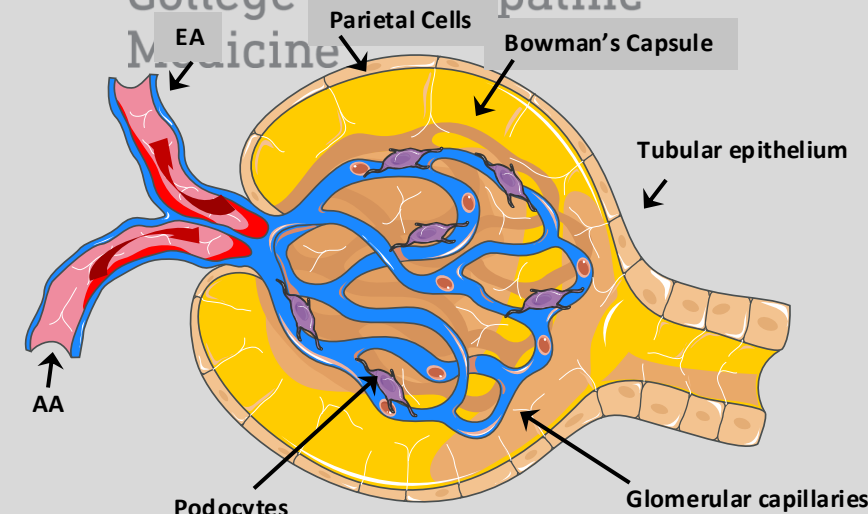
The Glomerulus & Bowman's Capsule

Development:

- The ureteric bud gives rise to the urinary system from the collecting duct to the ureters
- The tubular epithelium between Bowman's capsule & the connecting ducts arise from condensed mesenchymal cells.
- The proximal end of tubular epithelium attaches to the arterial vascular bundle that develops into the glomerular capillary tuft.
- Thinning of the epithelium on one blind end of tubule leads to the emergence of the parietal layer of Bowman's capsule.
- In contrast, the opposite visceral layer thickens and attaches to the glomerular capillaries.
- These visceral epithelial cells later fold and develop into podocytes.
- Podocytes:
 - Modified epithelial cells
 - Represent the visceral layer of Bowman's capsule.
 - Podocytes are continuous with the parietal layer of Bowman's capsule.
- Ultrafiltrate drains into Bowman's space and flows into the proximal tubule at the urinary pole of the renal corpuscle.

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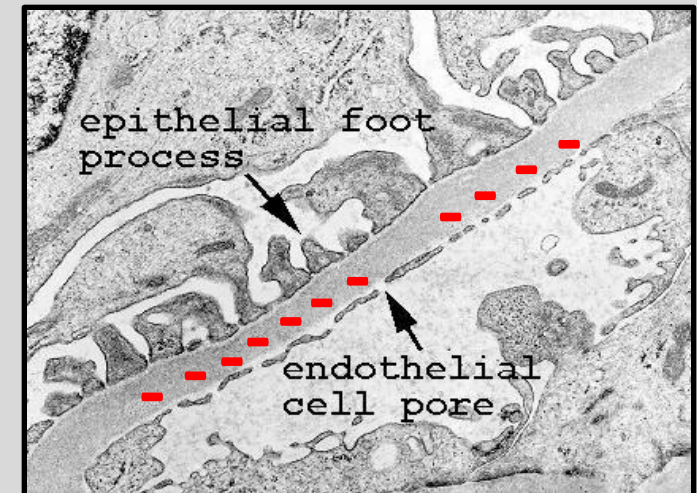
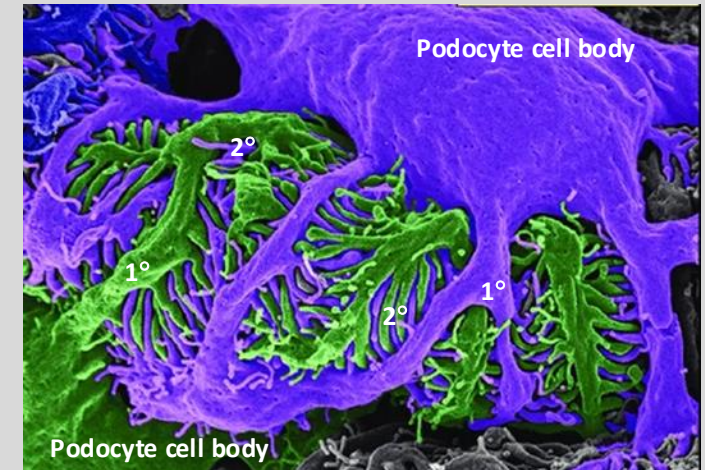


The Glomerular Filtration Barrier

- In the mature kidney, foot processes of the podocytes cover the glomerular capillaries.
- Glomerular filtration barrier is between the glomerular capillary lumen and Bowman's space
 - Glycocalyx:
 - Negatively charged glycosaminoglycans
 - Prevents leakage of large negatively charged macromolecules.
 - Endothelial cells:
 - Surrounded by the glomerular basement membrane and a layer of podocyte foot processes.
 - Contain large fenestrations, 70-nm holes
 - No restriction to movement of H_2O , small solutes, small proteins out of the capillary
 - Basement membrane:
 - Located between endothelial cells (inner) and podocyte (outer) foot processes
 - Three layers
 - Lamina rara interna: Inner thin layer
 - lamina densa: Thick layer
 - Lamina rara externa: Outer thin layer
 - Contains heparan sulfate proteoglycans
 - Restricts large, negatively charged solutes

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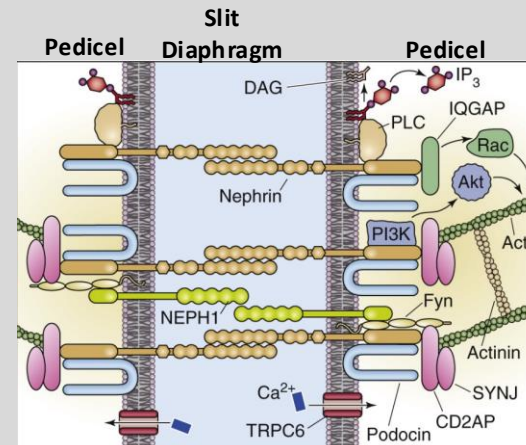
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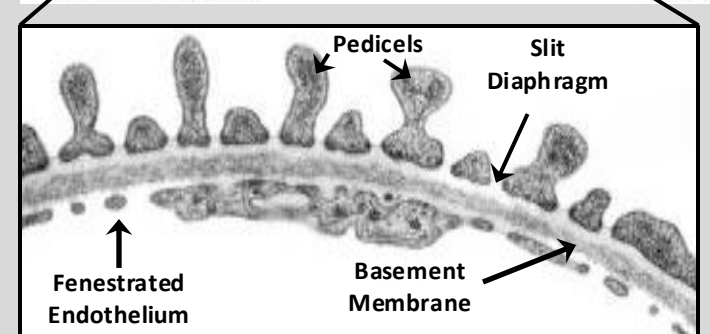
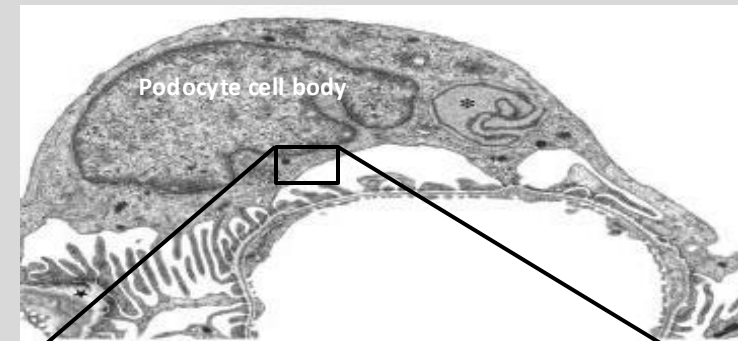
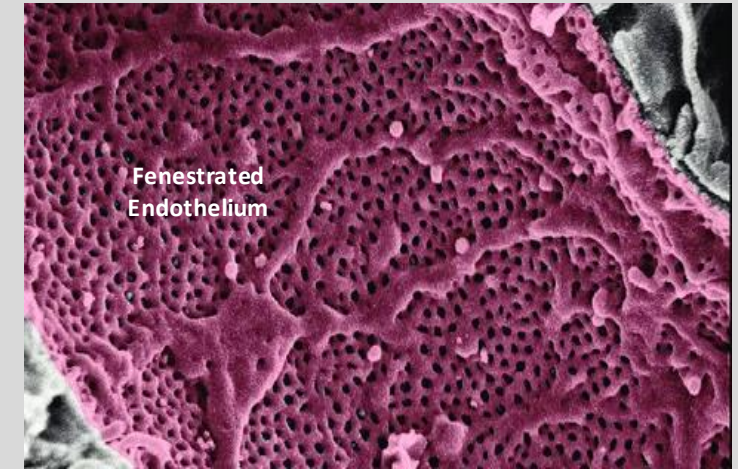
The Glomerular Filtration Barrier

Podocytes:

- Cell body links to the foot processes by leg-like primary and secondary processes.
- These foot processes (pedicels) interdigitate & cover the basement membrane.
- Between interdigitations are filtration slits & are connected by thin slit diaphragms (~ 4-14 nm).
- Glycoproteins with negative charges cover the podocyte cell body, interdigitations & the slit diaphragms.
- Contribute to the restriction of filtration of large anions.
- Slit diaphragm integrity is maintained by integral membrane proteins such as nephrin, NEPH1 (also known as Kirrel1) & podocin.
- Phosphotyrosine motifs on IC domains of these proteins to recruit signaling molecules controlling slit permeability.
- Defects in these proteins can make the filtration barrier leaky.
- Results in the appearance of large proteins (e.g., albumin) in the urine (e.g., nephrotic syndrome).



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The Glomerular Filtration Barrier

Mesangial cells:

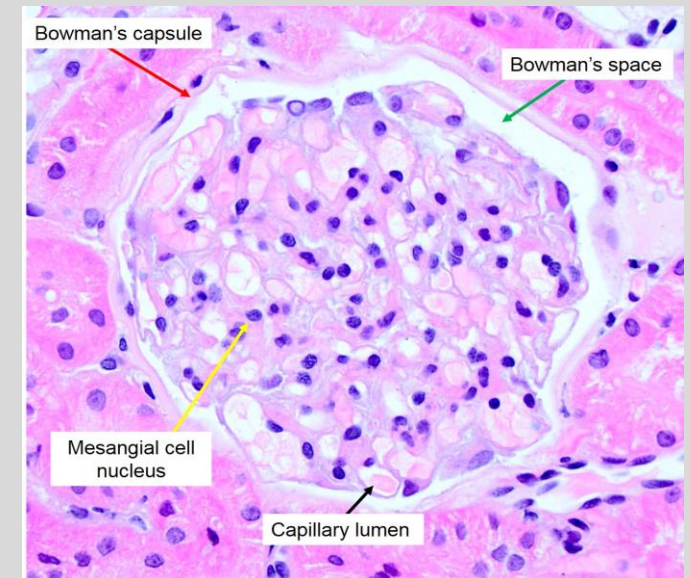
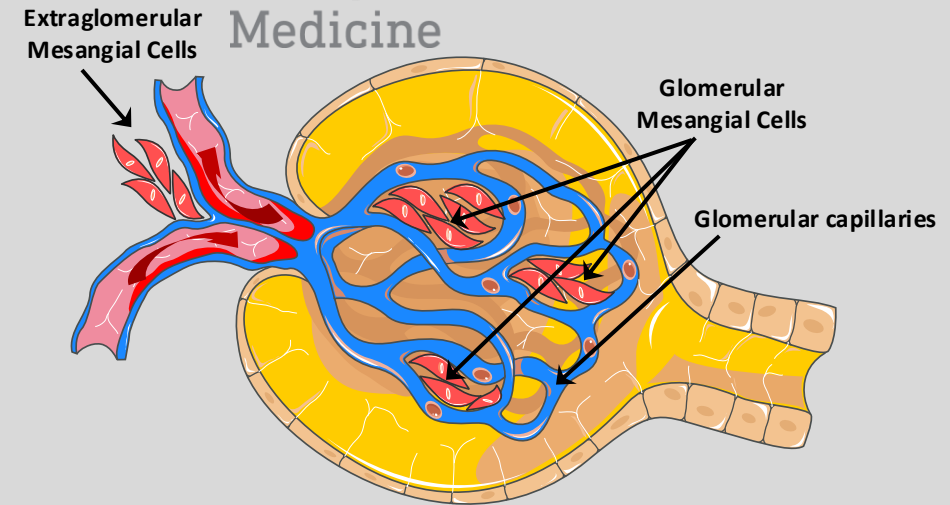
- Originate from embryonic mesenchymal cells
- Role is to maintain structural integrity of the glomerular microvasculature & ECM homeostasis.
- Possess contractile properties of smooth muscle cells.
- Influence glomerular filtration rate (GFR) by regulating blood flow through the glomerular capillaries or by altering the capillary surface area.
- Provide structural support for the glomerular capillaries
- Secrete extracellular matrix
- Exhibit phagocytic activity by removing macromolecules from the mesangium.
- Secrete prostaglandins & proinflammatory cytokines.

Extraglomerular Mesangial cells:

- Found outside the GL (between the afferent and efferent arterioles).
- Interact with *Macula Densa* (MD; specialized tubule epithelial cells) & granular cells of the afferent arterioles to form the Juxtaglomerular Apparatus (JGA).
- One component of the tubuloglomerular feedback (TGF) mechanism.

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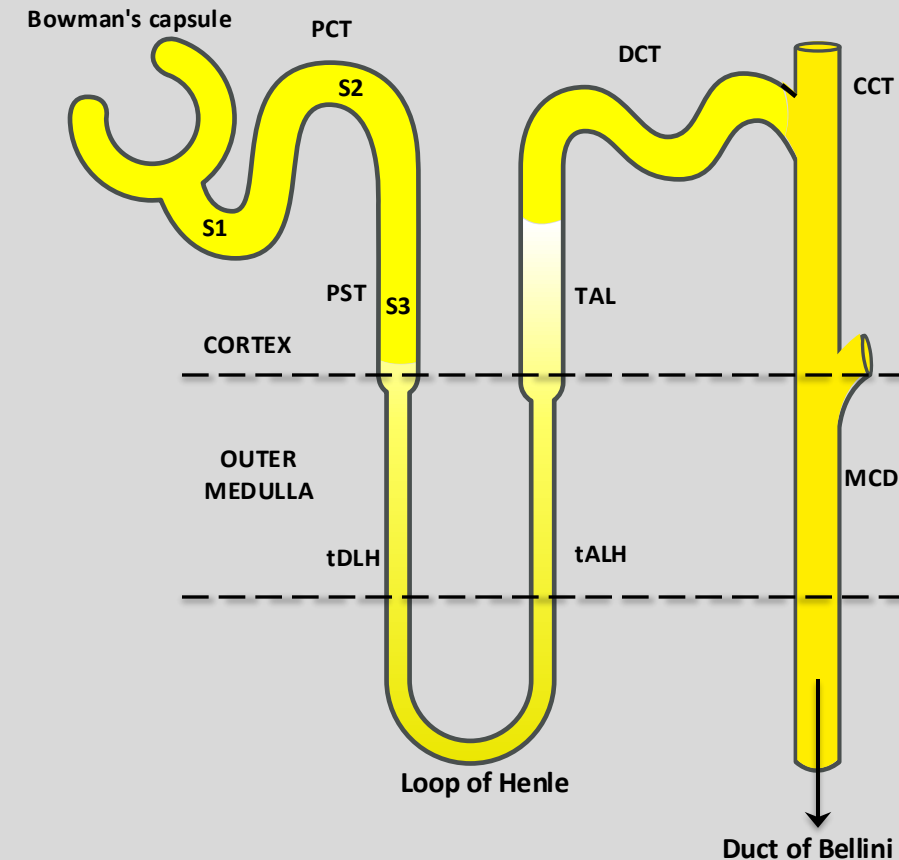


Tubular Components of the Nephron

- Embryologically derived from the ureteric bud & metanephric blastema.
- An epithelial structure designed to convert ultrafiltrate → urine.
- Bowman's capsule (glomerular capsule):
 - Surrounds the glomerulus & contains Bowman's space
 - Contiguous with the lumen of the tubule.
- Proximal tubules are subdivided into 3 segments based on their appearance:
 - Proximal Convoluted Tubule (PCT, S1 & S2):
 - Cells have apical microvilli and basolateral membrane (brush border)
 - Rich supply of mitochondria
 - Proximal straight tubule (PST, S2 & S3):
 - Cells have diminished ultrastructural complexity.
- Thin descending limb (tDLH) & thin ascending limb (tALH):
 - Less complex and flatter cells than PCT & PST
- Thick ascending limb of the loop of Henle (TAL):
 - Cells have tall interdigitations and numerous mitochondria within invaginated basolateral membranes
- Distal convoluted tubule (DCT):
 - Cells are similar to the thick ascending limb
- Cortical collecting tubule (CCT) & medullary collecting duct (MCD):
 - Composed of INTERCALATED and PRINCIPAL cells

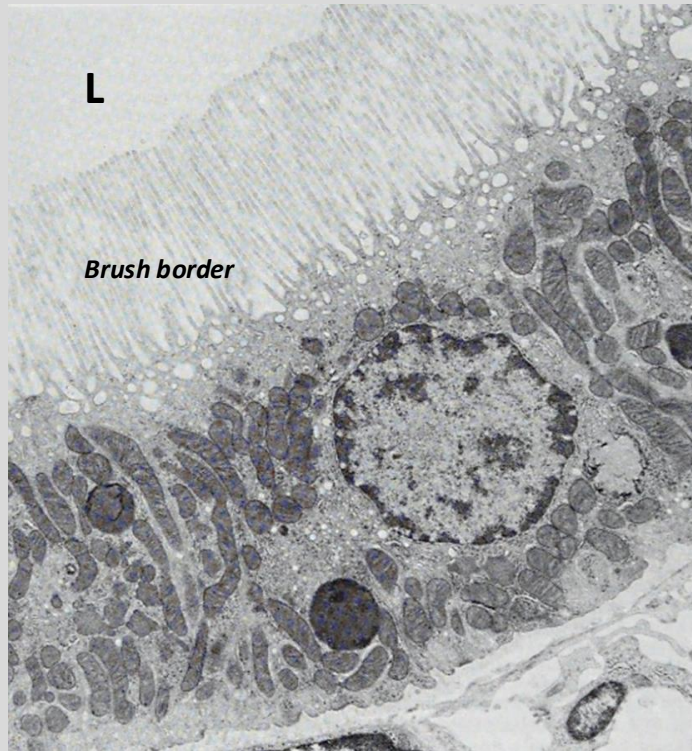
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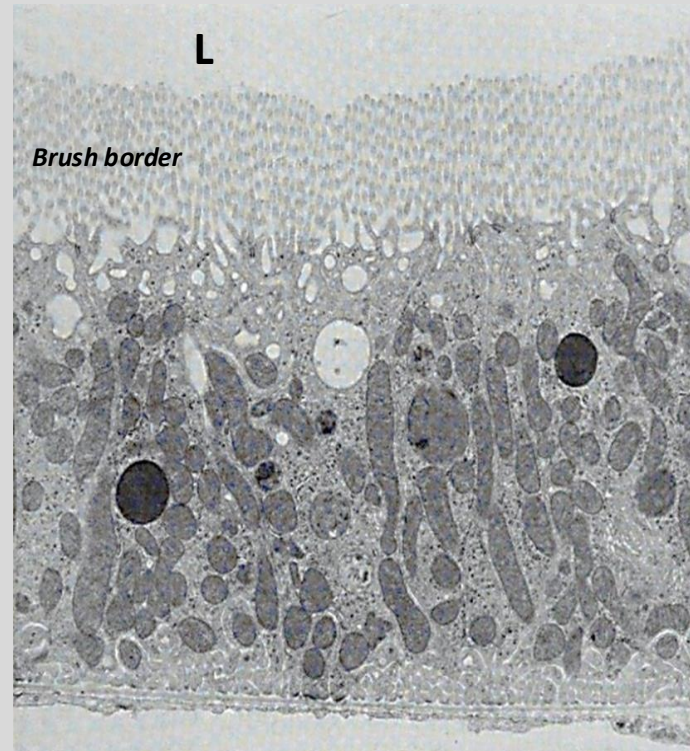


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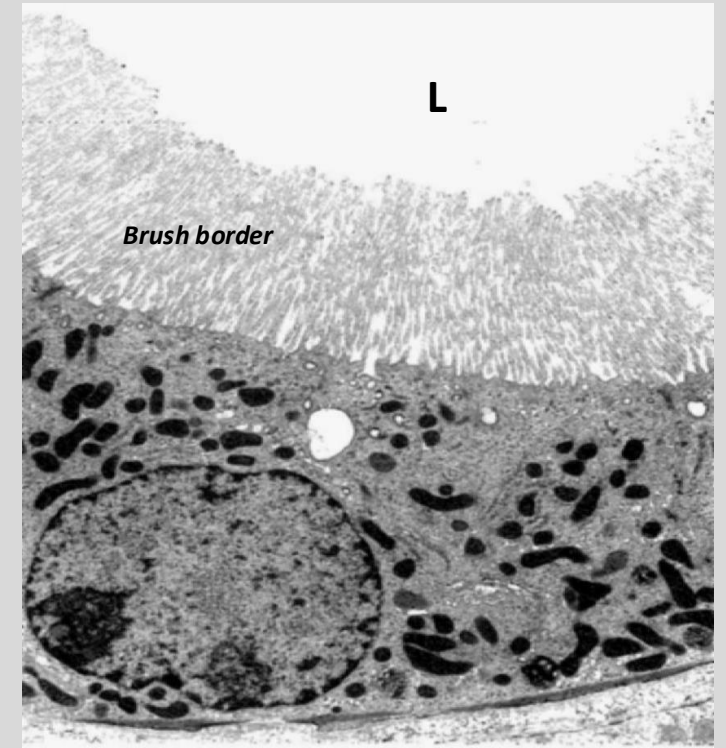
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PCT (S1)



PCT (S2)



PST (S3)

Tubular Components of the Nephron

Proximal tubule:

- Reabsorbs 67% of the filtered H_2O & electrolytes (isosmotic)
- Reabsorbs all glucose, proteins, AA & most of HCO_3^- in the ultrafiltrate
- Highly permeable to H_2O (due to expression of AQP1) in the apical & basolateral membranes

Thin descending limb of the Loop of Henle:

- Reabsorbs 15% of the filtered H_2O (mediated by AQP1).

Thin ascending limb of the Loop of Henle:

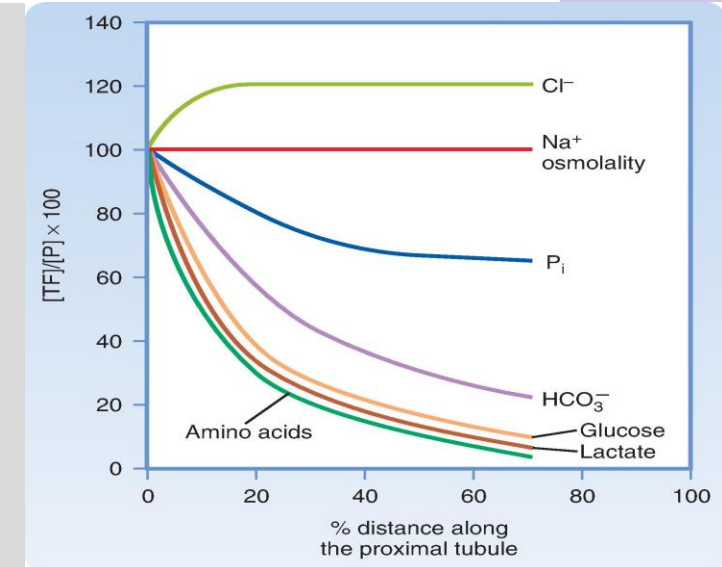
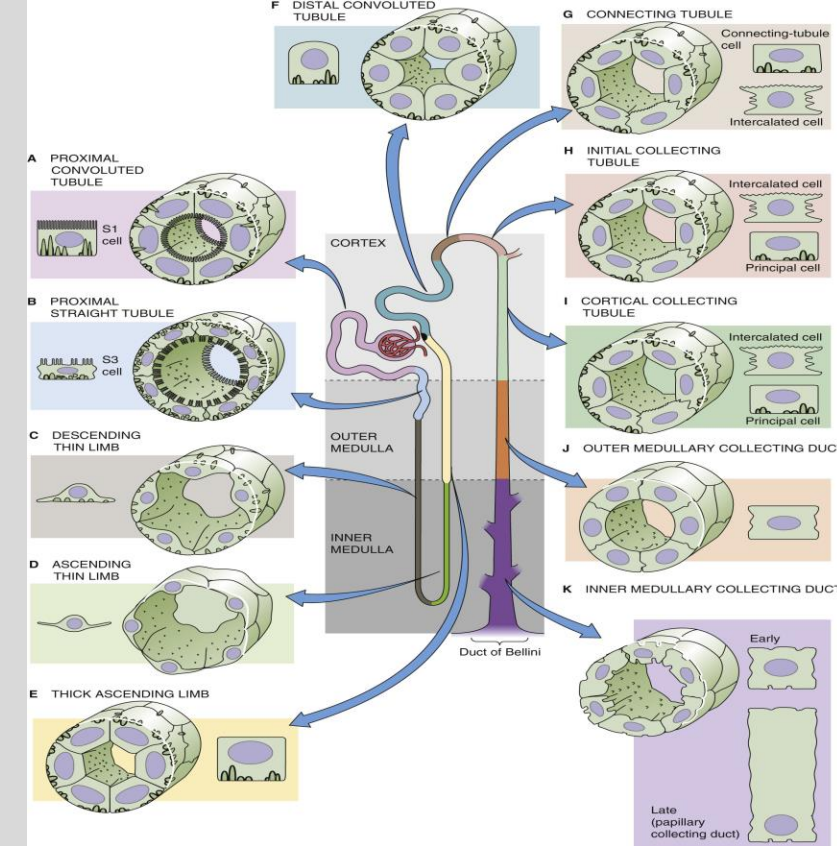
- Reabsorbs NaCl by a passive mechanism.
- H_2O reabsorption in the descending thin limb increases $[\text{NaCl}]$ in the tubule fluid.
- NaCl-rich fluid moves toward the cortex
- NaCl diffuses out of the tubule lumen & into the medullary interstitial fluid.

Thick ascending limb of the Loop of Henle:

- Impermeable to water.
- Reabsorbs 25% of filtered NaCl.
- K^+ is secreted & reabsorbed.
- Mg^{++} & Ca^{++} are reabsorbed paracellularly.
- Remainder of HCO_3^- is reabsorbed.

Distal convoluted tubule & Collecting ducts:

- Reabsorbs ~ 8% of the filtered NaCl
- Secrete variable amounts of K^+ & H^+
- Reabsorbs $\approx 8\text{-}17\%$ of H_2O .



H₂O & Solute Permeability in the Nephron

- Kidneys produce a range of urine osmolalities (concentrated (high mOsm)→ dilute (low mOsm)).
- Nephron segments have differential permeabilities to solute & H₂O.

Proximal Tubule:

- H₂O reabsorption: Paracellular
- Solute reabsorption: Transcellular & Paracellular

Thin Descending Limb of the Loop of Henle:

- H₂O reabsorption: Paracellular
- Solute reabsorption: None

Thin Ascending Limb of the Loop of Henle:

- H₂O reabsorption: None
- Solute reabsorption: Paracellular

Thick Ascending Limb of the Loop of Henle:

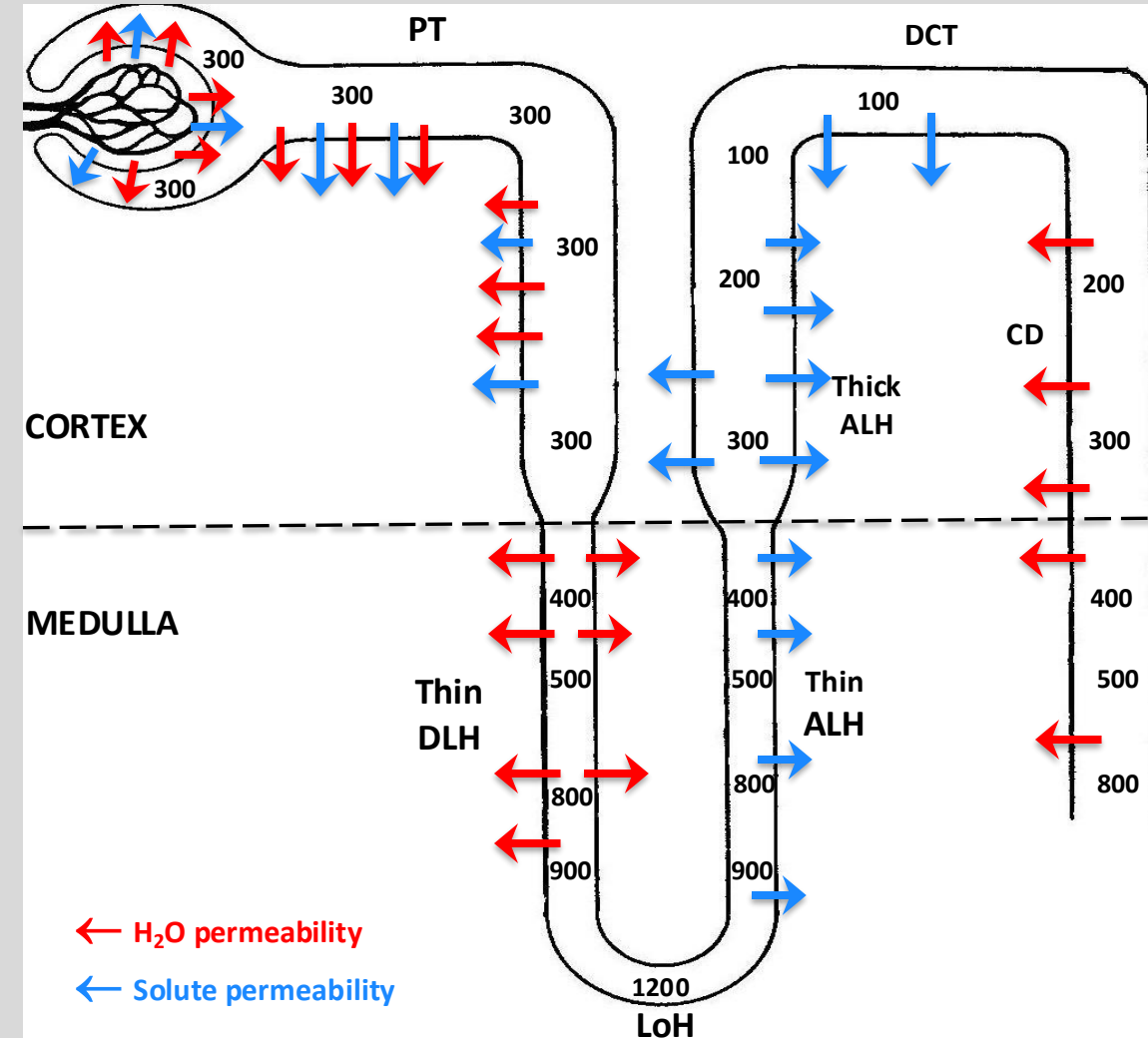
- H₂O reabsorption: None
- Solute reabsorption: Transcellular & Paracellular

Distal Tubule:

- H₂O reabsorption: None
- Solute reabsorption: Paracellular

Collecting Duct:

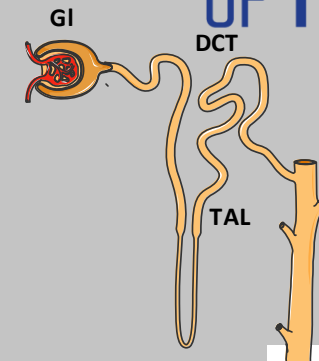
- H₂O reabsorption: Transcellular
- Solute reabsorption: Transcellular & Paracellular



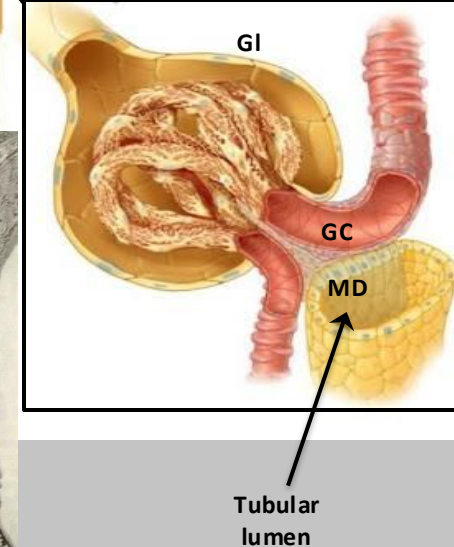
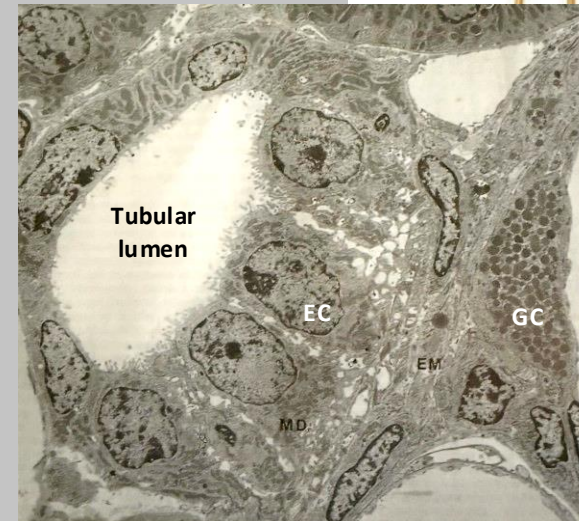
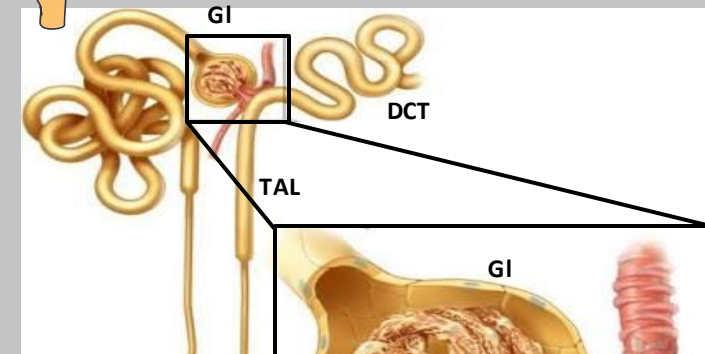
The Juxtaglomerular Apparatus

- Consists of extraglomerular mesangial cells, the *macula densa*, and granular cells (GC).
- *Macula densa*:
 - A patch of specialized tubule epithelial cells (EC) at the transition between the TAL and DCT
 - Contacts its “own” Gl.
 - Cells have large nuclei and are closely packed
- Granular cells (GC) of the afferent arterioles (juxtaglomerular or epithelioid cells) are specialized smooth-muscle cells that produce, store, and release the enzyme renin.
- JGA is part of a feedback mechanism that directly regulates RBF and filtration rate & indirectly modulates Na^+ balance & systemic BP.
- Two regulatory roles:
 - Tubuloglomerular feedback (TGF): Regulation of GFR depending on $\text{H}_2\text{O}/\text{NaCl}$ reaching a *macula densa* increases.
 - Control of RBF: Detects changes in the pressure of the renal artery via baroreceptors in AA, directing granular cells to modulate renin (rē-nən; ree-nen) release into circulation and regulating the renin-angiotensin-aldosterone system (RAAS) .

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Macula (spot) densa (dense)



Renal Innervation

Efferent Innervation:

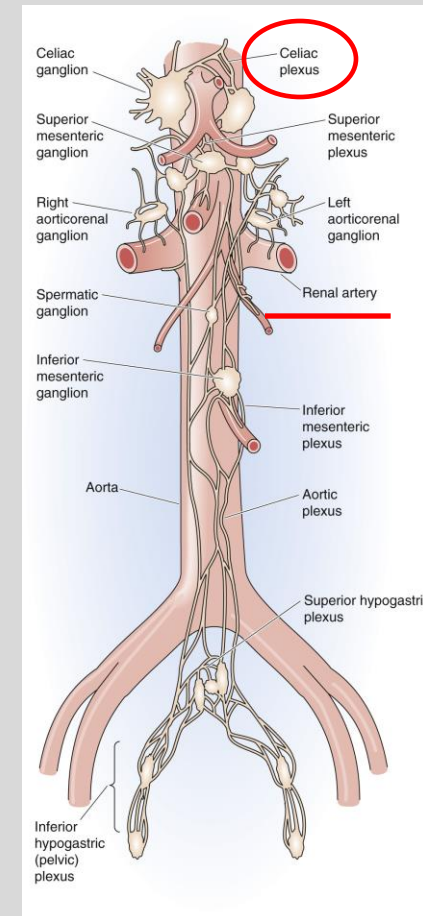
- Innervated by the sympathetic nervous system (SNS).
- One of the organs that has single innervation (Sweat glands, circulatory system, adrenal medulla, pancreas, liver, adipose tissues, kidney)
- Lack parasympathetic innervation.
- Renal SNS originates from the celiac plexus & follows the arterial vessels into the kidney
- Releases norepinephrine (NE) into the loose connective tissue near the smooth muscle cells of the vasculature (i.e., renal artery, AA, EA) & the proximal tubules.
- 3 major effects:
 - Vasoconstriction.
 - Enhance Na^+ reabsorption by proximal tubule cells.
 - Stimulate renin secretion (GC of the AA).

Afferent Innervation

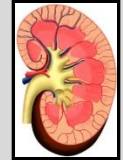
- Renal nerves also include afferent (i.e., sensory) fibers
- Detects baroreceptor and chemoreceptor impulses.
- Baroreceptors: Detect changes in perfusion pressure in the interlobular arteries & AA.
- Chemoreceptors: Detect elevated E.C. $[\text{K}^+]$ & $[\text{H}^+]$ in the renal pelvis

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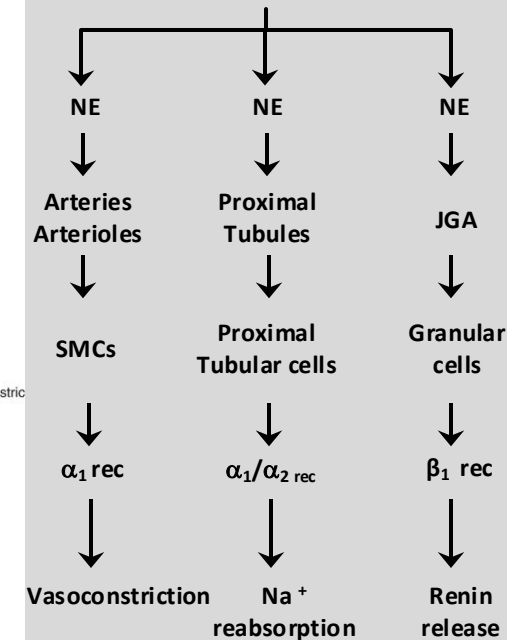
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Renal Efferents



Sympathetic Nervous system



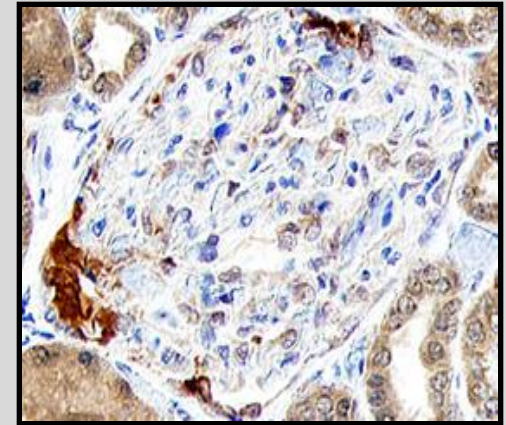
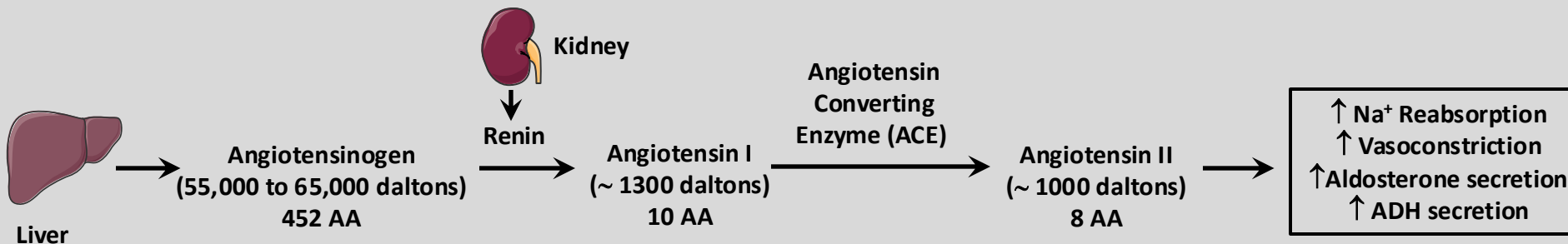
Note: Intrarenal production of dopamine is not regulated by renal sympathetic nerve activity,

The Endocrine Kidney

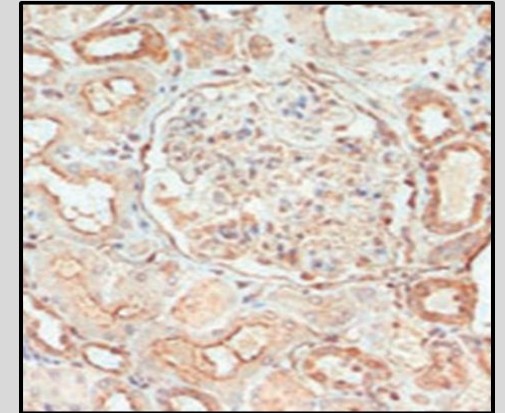
- The kidneys, as endocrine organs, produce renin, 1,25-dihydroxyvitamin D & erythropoietin (Epo).
- Also produce prostaglandins & bradykinin as paracrine agents that control circulation.
- **ENDOCRINE:** Product of an organ circulating in the body and producing an effect remote from its point of origin
- **PARACRINE:** Product secreted by a cell and acting on adjacent cells
- **AUTOCRINE:** Product secreted by a cell and acting on the same cell

Renin:

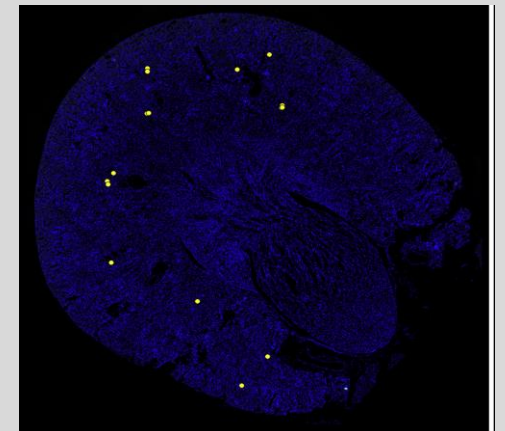
- Enzyme that initiates the cascade generating angiotensin peptides (a.k.a. Angiotensinogenase)
- ~ 30-50 Kd plasma protein
- Regulate blood pressure, cell growth, apoptosis & electrolyte balance.
- Rate limiting enzyme of the renin-angiotensin-aldosterone system (RAAS).
- Angiotensin II: Main effector peptide



Renin



CYP27B1

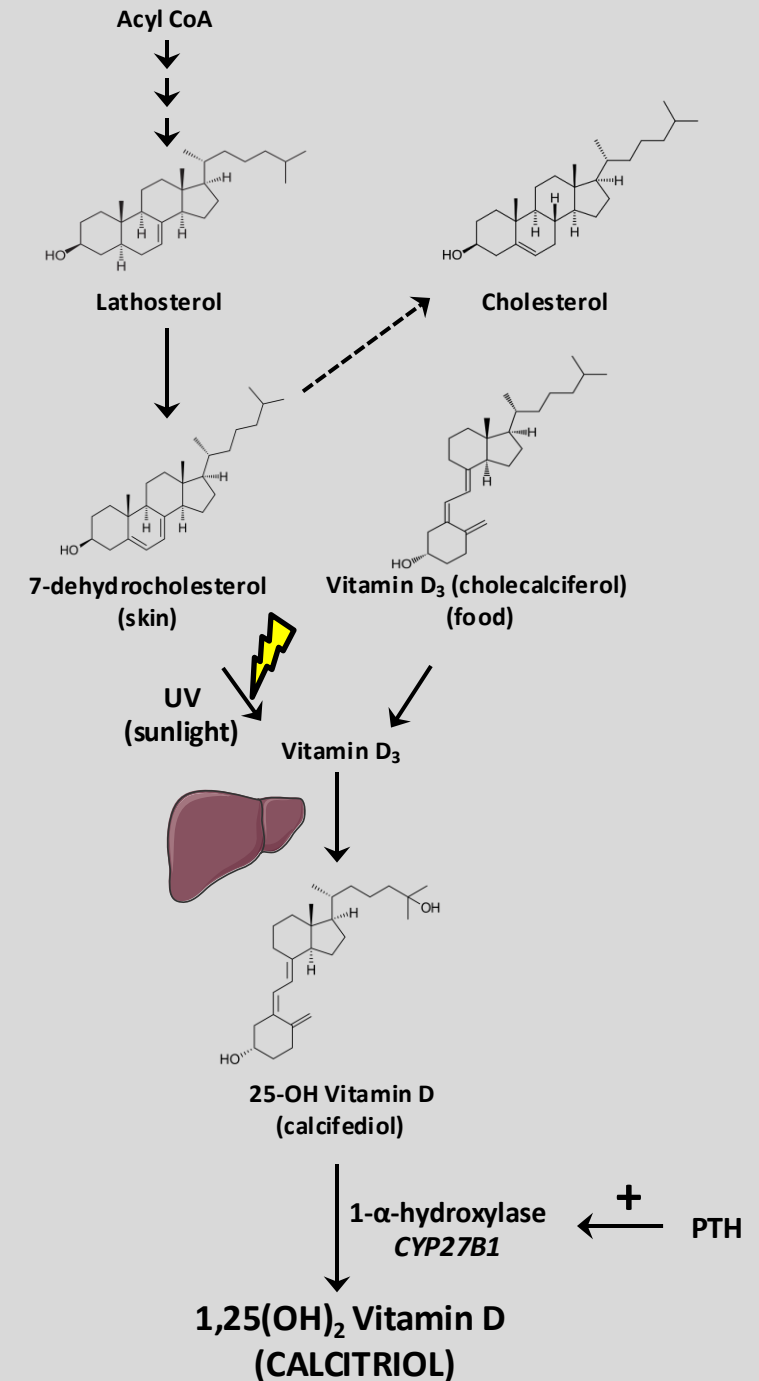


Epo

The Endocrine Kidney

VITAMIN D:

- Not technically a vitamin
- A fat-soluble secosteroid hormone.
- Promotes Ca^{2+} conservation in the body by increasing intestinal Ca^{2+} absorption and reducing urinary Ca^{2+} loss.
- Steroid-derived precursors are ingested or produced by the action of UV on the skin.
- Active form of vitamin D is 1,25-dihydroxycholecalciferol ($1,25\text{-(OH)}_2$ vit D).
- Liver produces 25-hydroxycholecalciferol (25-OH vit D_3).
- 25-OH vit D_3 is converted to $1,25\text{-(OH)}_2$ vitamin D in the kidney.
- Synthesis occurs in proximal tubules.
- Conversion is under the control of parathyroid hormone (PTH).
- Attaches to vitamin D receptors (VDR)
- VDR:
 - Nuclear receptors
 - Found in various tissues throughout the body.



The Endocrine Kidney

ERYTHROPOIETIN (EPO):

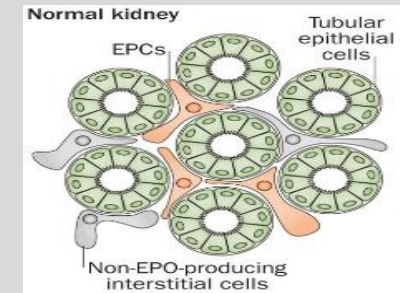
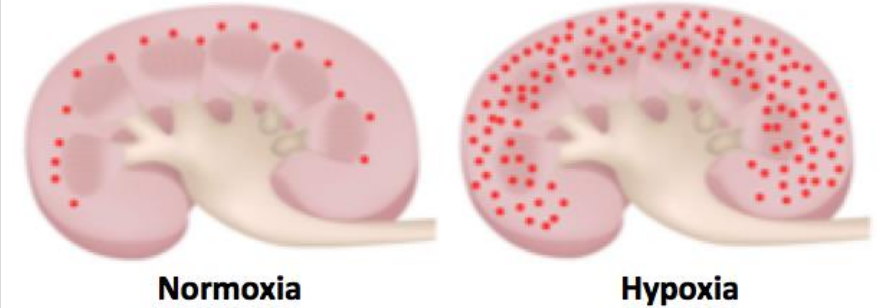
- Most important regulator of the proliferation of erythroid progenitors.
- Stimulates proliferation and maturation of committed erythroid progenitors to increase RBC production in the bone marrow.
- First human hematopoietic growth factor to be isolated.
- 34–39 kDa glycoprotein.

In adults :

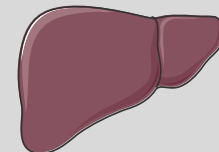
- 90% produced in kidneys (Erythropoietin Producing Cells (EPCs), Peritubular Interstitial cells)
- 10% in liver (perisinusoidal cells)
- Fibroblast-like cells in the interstitium of the cortex and outer medulla secrete EPO in response to a fall in the local tissue oxygen content (P_{O_2}).

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Number of Cells in Kidney Producing Erythropoietin

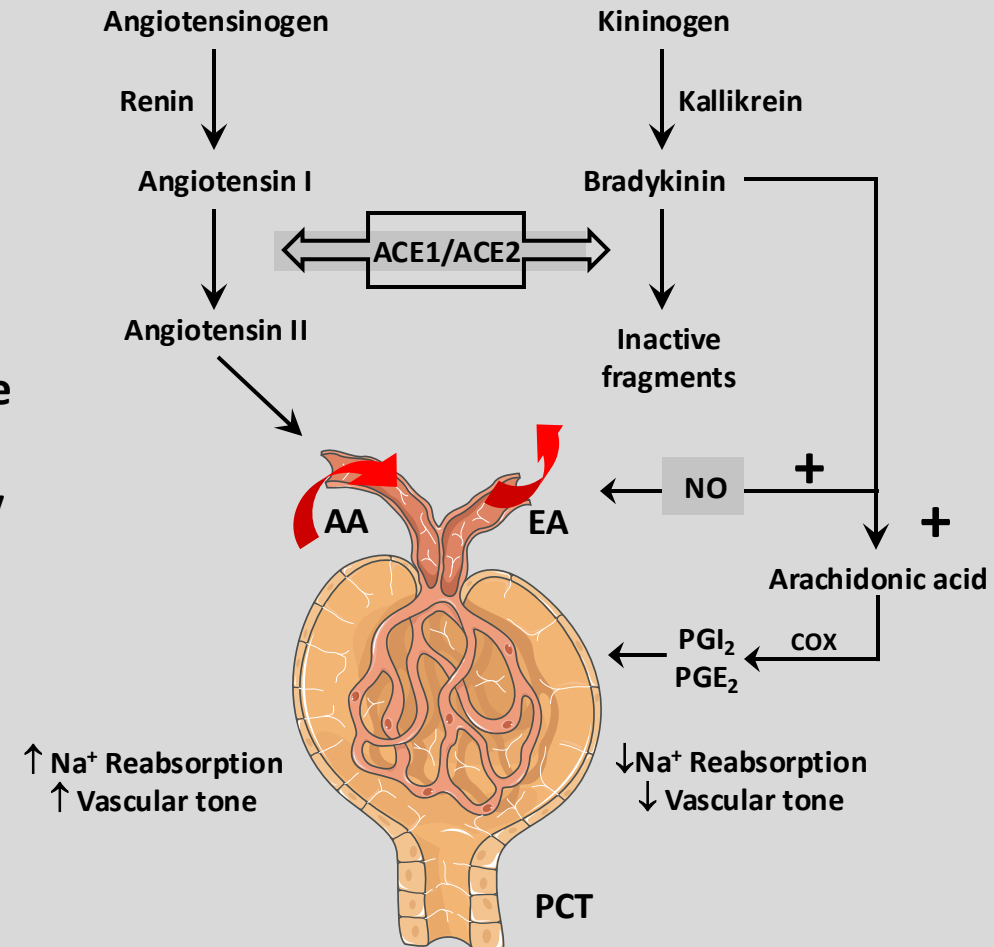


Plasma Epo



The Paracrine Kidney

- Bradykinin is a potent part of the system that induces hypotension and vasodilation.
- Degraded by angiotensin-converting enzyme (ACE)
- The kidney releases bradykinin & vasodilator prostaglandins (PGI_2 , PGE_2)
- These substances are generally vasodilators & play a protective role when renal blood flow is compromised.
- Bradykinin acts directly on the tubular cells in the nephron to reduce Na^+ reabsorption.
- Bradykinin also stimulates nitric oxide (NO) production in vascular endothelial cells (EC) and release of arachidonic acid (AA) from membrane PLs in VSMC and ECs.
- Arachidonic acid is then metabolized to vasodilator and natriuretic PGs by the enzyme cyclooxygenase (COX).

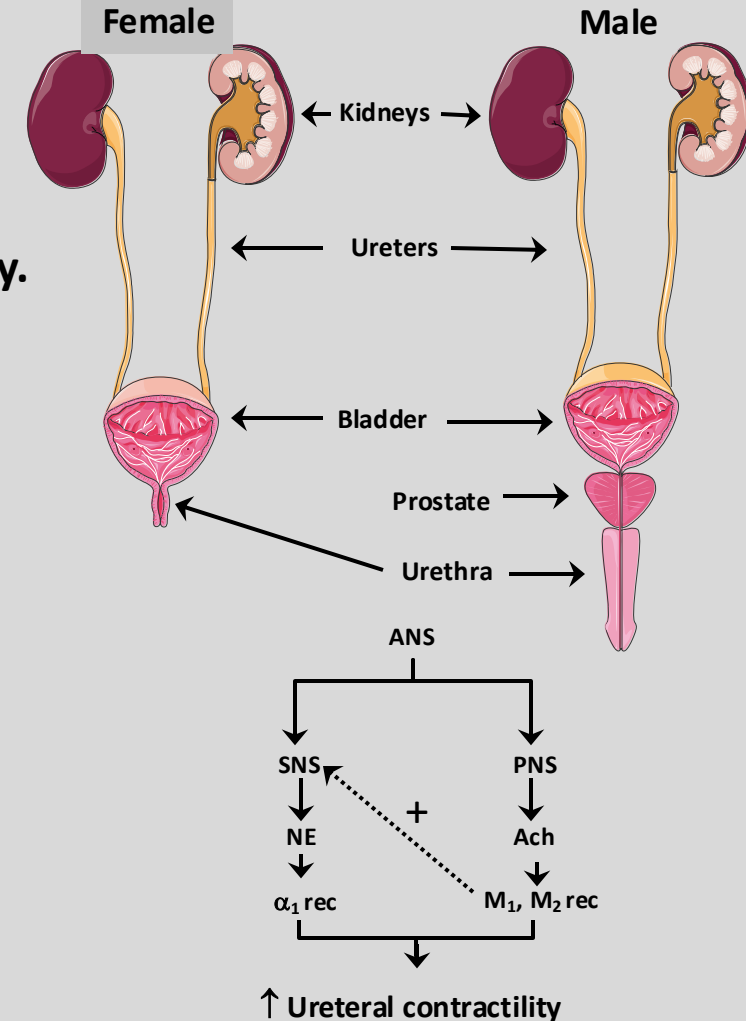


The Ureters and Bladder

- By the time the fluid leaves the most distal portion of the collecting duct, the filtrate has the final composition of urine.
- Renal pelvis, ureters, bladder, and urethra do not modify the urine volume or composition
- Ureters:
 - Serve as conduits for the passage of urine from the renal pelvis → bladder
 - Each ureter is located in the retroperitoneum
 - Loop over the top of the common iliac artery & vein on the same side of the body.
- The lumen of each ureter is lined:
 - Transitional epithelium
 - Submucosal layer of connective tissue
 - Inner longitudinal & outer circular layer of smooth muscle.
- Ureteral smooth muscle functions as a syncytium (e.g., unitary smooth muscle).
- Gap junctions conduct electrical activity from cell to cell.
- Ureteral peristaltic waves:
 - Originate from electrical pacemakers in the proximal portion of the renal pelvis.
 - ANS can modulate peristalsis, but it's not an absolute requirement.
- SNS (via aortic, hypogastric, and ovarian or spermatic plexuses) & PNS input enhance ureteral contractility.

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The Ureters and Bladder

- Main portion (body) that collects urine and a funnel-shaped extension (neck) that connects with the urethra
- The bladder lumen is lined by transitional epithelium.
- Bladder wall: 3 poorly-defined layers of SM (detrusor muscle) & lack gap junctions.
- At the lower tip of the trigone, the bladder opens into the posterior urethra.
- Posterior urethral wall: Smooth-muscle fibers of the detrusor muscle interspersed with elastic tissue, forming the Internal Sphincter (IS).
- Adjacent is the External Sphincter (ES), made up of voluntary, slow-twitch striated-muscle fibers.

SNS innervation

- Origin: Intermediolateral cell column of T10 – L2.

Preganglionic fibers:

- Lumbar splanchnic nerves → superior hypogastric plexus → L&R hypogastric nerves → Inferior hypogastric/pelvic plexus → postganglionic fibers (to the bladder wall via the distal portion of the hypogastric nerve).

PNS innervation:

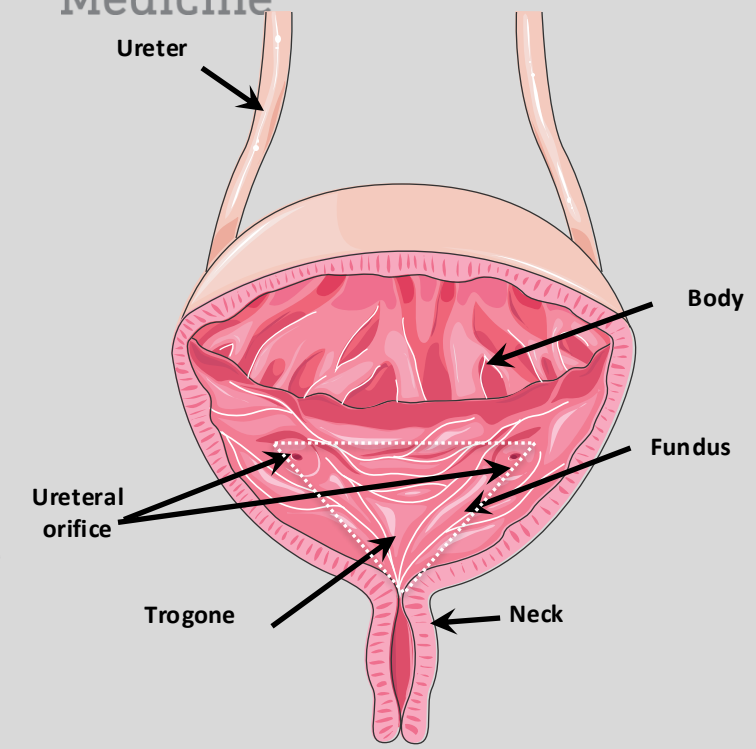
- Origin: Intermediolateral cell column of S2-S4

Preganglionic fibers:

- Pelvic splanchnic nerve → postganglionic neurons (body & neck)

Somatic innervation:

- Origin: Motor neurons (S2 – S4) → Pudendal nerve



	IS	ES
Muscle type	Smooth	Skeletal
Innervation	Autonomic	Somatic
Nerve	Hypogastric	Pudendal

Control of Bladder Tone

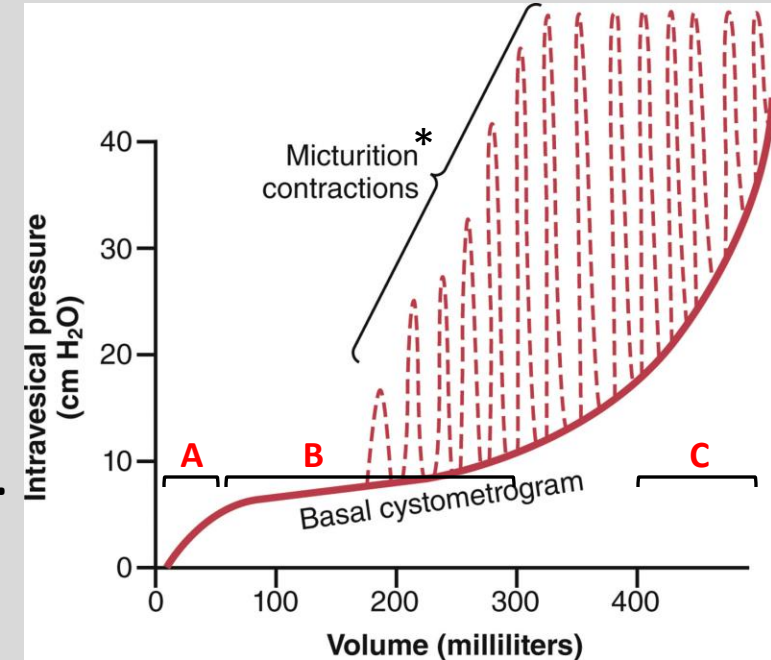
- Bladder tone is defined by the relationship between bladder volume & internal (intravesical) pressure & can be evaluated by a cystometrogram.

A: Increase in bladder volume from 0~50 mL produces a moderately steep increase in pressure.

B: Volume increases (to ~300 mL) produce almost no pressure increase, reflecting relaxation of bladder smooth muscle.

C: At volumes >400 mL, additional increases in volume produce steep increments in pressure. Bladder tone, up to the point of triggering the micturition reflex, is independent of extrinsic bladder innervation.

- The micturition reflex is normally inhibited by the pontine micturition center (PMC).
- The urge for voluntary bladder emptying starts ~150 mL
- Senses fullness at 400-500 mL
- Voiding phase:
 - Begins with a voluntary relaxation of the ES followed by relaxation of the IS
 - Afferents signal are sent when a small amount of urine reaches the proximal (posterior) urethra.
 - The micturition reflex: the PMC no longer inhibit the parasympathetic preganglionic neurons that innervate the detrusor muscle.
 - Result: The bladder contracts, expelling urine.
 - The initial bladder contractions → Sensory impulses from stretch receptors & establishing a self-regenerating process

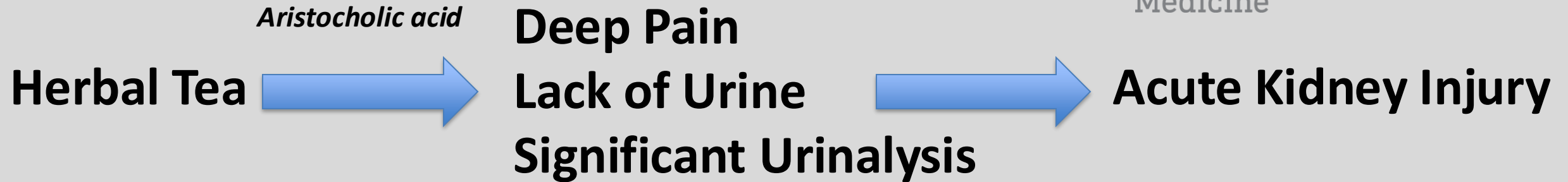


Cystometrogram (CMG):

- Diagnostic test that measures bladder pressure, capacity & muscle function
- Via insertion of a catheter through the urethra & emptying the bladder, and then recording the pressure while filling the bladder with 50-mL increments of water

*Micturition = urination

PMC: Cortical and suprapontine centers in the brain



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Case Report

Acute Kidney Injury Following the Ingestion of a Medicinal Plants' Mixture: A Case Report

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Acute renal failure secondary to medicinal plants is common in countries where the use of traditional phytotherapy is preponderant. Although the nephrotoxic potentials of some herbal preparations have been well characterized, the use of many medicinal plants is still considered largely safe, often relying on weak evidence. Here, we report the case of a 17-year-old patient with severe acute renal failure, associated to an esophagitis with erosive gastritis as well as an inflammatory anemia, with no obvious etiology. After ruling out any other plausible explanation, the syndrome was attributed to the chronic intake of a mixture of three medicinal plants, previously unknown to be nephrotoxic: *Artemisia absinthium*, *Marrubium vulgare*, and *Centaurium erythraea*. A histological examination of a renal biopsy sample revealed an aspect of interstitial nephritis without antibody deposits. To our knowledge, this is the first reported case of acute kidney injury related to the consumption of these three plants and prompts further studies to carefully assess the safety of traditional medicinal products based on these plants.

Keywords: acute kidney injury; *Artemisia absinthium*; *Centaurium erythraea*; *Marrubium vulgare*; tubulointerstitial nephritis

Artemisia absinthium
Common wormwood



Marrubium vulgare
Common horehound



Centaurium erythraea
European centaury



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